Task 1: Posts Screen with Loading and Error States

```
import Foundation

struct Post: Decodable {
    let id: Int
    let title: String
    let body: String
}
let json = """
[
    {
        "id": 1,
        "title": "Swift Networking",
        "body": "Learn how to fetch API data"
    },
    {
        "id": 2,
        "title": "JSON Parsing",
        "body": "Decode JSON into Swift structures"
    }
]
"""
do {
    let data = Data(json.utf8)
    let posts = try JSONDecoder().decode([Post].self, from: data)
```

```
    print("Posts Loaded Successfully")
    for post in posts {
        print("\nID: \(post.id)")
        print("Title: \(post.title)")
        print("Body: \(post.body)")
    }
} catch {
    print("Error decoding JSON: \(error)")
}
```

Output




```
Posts Loaded Successfully

ID: 1
Title: Swift Networking
Body: Learn how to fetch API data

ID: 2
Title: JSON Parsing
Body: Decode JSON into Swift structures
```

Task 2: Pull to Refresh and Search

```
import Foundation
struct Post: Decodable {
    let id: Int
    let title: String
    let body: String
}
let json = """
[
    {"id": 1, "title": "Swift Basics", "body": "Learn Swift programming"},
    {"id": 2, "title": "UIKit Guide", "body": "Build iOS interfaces"},
    {"id": 3, "title": "Swift Networking", "body": "Fetch API data"}
]
"""
func loadPosts() -> [Post] {
    do {
        let data = Data(json.utf8)
        return try JSONDecoder().decode([Post].self, from: data)
    } catch {
        print("Error loading posts")
        return []
    }
}
let posts = loadPosts()
let searchText = "swift"
```

```
let filteredPosts = posts.filter {
    $0.title.localizedCaseInsensitiveContains(searchText)
}
print("Posts loaded: \(posts.count)")
print("Search: \(searchText)")
if filteredPosts.isEmpty {
    print("No results")
} else {
    for post in filteredPosts {
        print("\(post.id). \(post.title)")
        print(post.body)
    }
}
print("Refresh completed")
let refreshedPosts = loadPosts()
print("Posts after refresh: \(refreshedPosts.count)")
```

Output




```
Posts loaded: 3
Search: swift
1. Swift Basics
Learn Swift programming
3. Swift Networking
Fetch API data
Refresh completed
Posts after refresh: 3
```

Task 3: Empty and Error States in Swift

```
import Foundation
struct Post: Decodable {
    let id: Int
    let title: String
    let body: String
}

let json = """
[
    {"id": 1, "title": "Swift Basics", "body": "Learn Swift"},
    {"id": 2, "title": "UIKit Guide", "body": "Build interfaces"}
]
"""

do {
    let data = Data(json.utf8)
    let posts = try JSONDecoder().decode([Post].self, from: data)
    let searchText = "Python"

    let results = posts.filter {
        $0.title.localizedCaseInsensitiveContains(searchText)
    }
```

```
    if results.isEmpty {
        print("No posts found for: \(searchText)")
    } else {
        for post in results {
            print("\(post.id): \(post.title)")
            print(post.body)
        }
    }
} catch {
    print("Error: Unable to decode posts")
}
```

Output




```
No posts found for: Python
```